Strumii: An AI-Driven Approach to Personalized Guitar Learning

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Introduction

Artificial Intelligence is becoming more and more present in our day-to-day lives, but could it be useful in music education? Through our research, we want to determine if users are comfortable with an Artificial Intelligence teacher that would tailor guitar lessons to match the user's learning preferences and skill level. There are a multitude of challenges when trying to coordinate with a human guitar teacher, from scheduling conflicts to the financial burden that private lessons can create. We hope to observe future implications of how accessible musical learning and learning in the broader context of education can change with AI.

Overall, our research will contribute to insight into the population's perspective and opinions toward Artificial Intelligence as a teaching tool. This will be highly relevant to any future work in the development of the education system as well as how humans will interact with AI in a teaching role. Specifically, we aim to answer the question: Does AI help students achieve their learning goals and help them better understand assigned topics?

The web app *Strumii* aims to facilitate two of the most common challenges faced by people who wish to learn an instrument: money and time. Strumii's purpose is to minimize the financial and time barriers through personalized guitar lessons accessible to anyone with internet access. There is no payment or subscription required, just a motivation to learn and improve.

Our project addresses these issues by using an AI model to generate lessons at the request of the user. A user has agency in providing feedback on these lessons, generating a lesson more suited to their situation. Many existing guitar apps lack the option for feedback and hide their lessons behind paywalls, which can discourage users from continuing learning. In addition, Strumii was designed to accompany users' musical journey at their own pace through short-form lessons. The user can decide when they are ready to complete or repeat a lesson.

Our application will recognize when a user plays the notes of a chord correctly by using a pitch detector and comparing the frequency of the played string with the correct frequency. Then, the detector will verify if a chord is played correctly with every note played. As the user continues lessons, they will be assessed through more difficult chord progressions. Throughout our study, we are interested in seeing if AI-generated lessons help users achieve their musical goals and enhance their motivation

Background

There were three applications that shared the technology and implementation of AI in similar ways we envision our app to function. While the applications had the same structure, they all focused on different aspects of learning how to play music, so we did not find direct comparisons.

Chordie AI is a mobile app that can generate chords for popular music from an existing list that the user can search. The app teaches all the chords to be able to play a specific song chosen by a user. There is a simple chat-bot to ask questions about music or the song you are learning. The application's main selling point is being able to teach you popular songs without having to learn music theory.

Issues come in with the integration of highly animated UI to gamify the learning process. The user must endure very slow and repetitive animations when learning new chords. There is no way to skip the long lessons to "learn" a chord or an option to say that you know how to play that chord, leading to frustration from intermediate players who may already have some guitar knowledge. Moreover, Premium users get access to the full music library, a better chord recognizer model, and "Unlimited Tailored Lessons," which are largely invisible to a user (like me) who does not/ cannot pay.

FretPro is another mobile app where a user utilizes their phone's microphone in order to learn & memorize the notes on the fretboard. The app will call out a note and a string, and the user needs to play it as quickly as they can. We can use inspiration from this app in order to implement other memorization tools for the user. Another website, musictheory.net, helps users visually learn music notes on a staff, but also has categories for their practice modules, like choosing the scale, difficulty, clef, etc. MusicTheory aims to focus on growing the user's understanding of theory in general, and serve a good basis in our app for lessons that do not require an instrument. While both of these applications do not use AI in their main goals, they serve as a starting point and reference to how our app should approach the structure and flow of lessons. It will be useful to see how lessons have been implemented and what makes those lessons work and what does not by testing and seeing what we would like to include in our application.

Youcisian has a free guitar mobile app that aims to teach the user how to play the guitar by gamifying learning through the use of audio recognition to assess your playing. The app has a tuner and exercises for users to get comfortable with playing the guitar before actual lessons. The main issue with this application is the lack of theory practice and rhythm focus, as you can play songs in the wrong rhythm and as long as you get the notes the listener is looking for, you are playing the songs perfectly. I did not see any explicit use of AI in their app. I found articles relating to music education and the integration of AI into that field to more completely understand why this app would or could help our target audience learn music effectively. These articles could also give us insight into the reception of artificial intelligence in music education, providing insight and clarity to where it's headed.

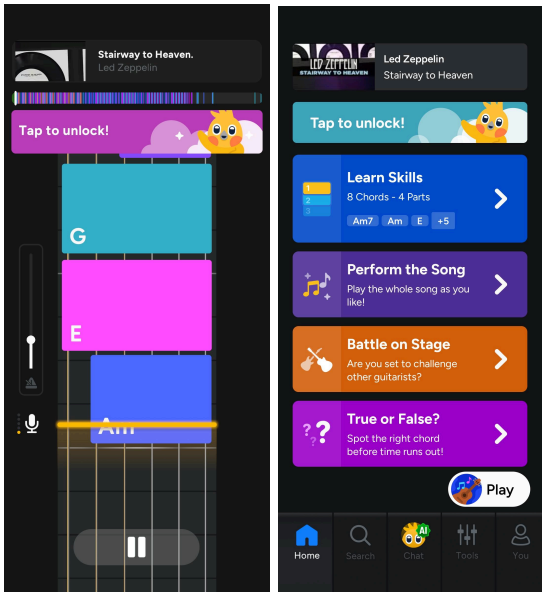

Figure 1: ChordieAI's User Interface for Lessons and Homescreen, mobile.

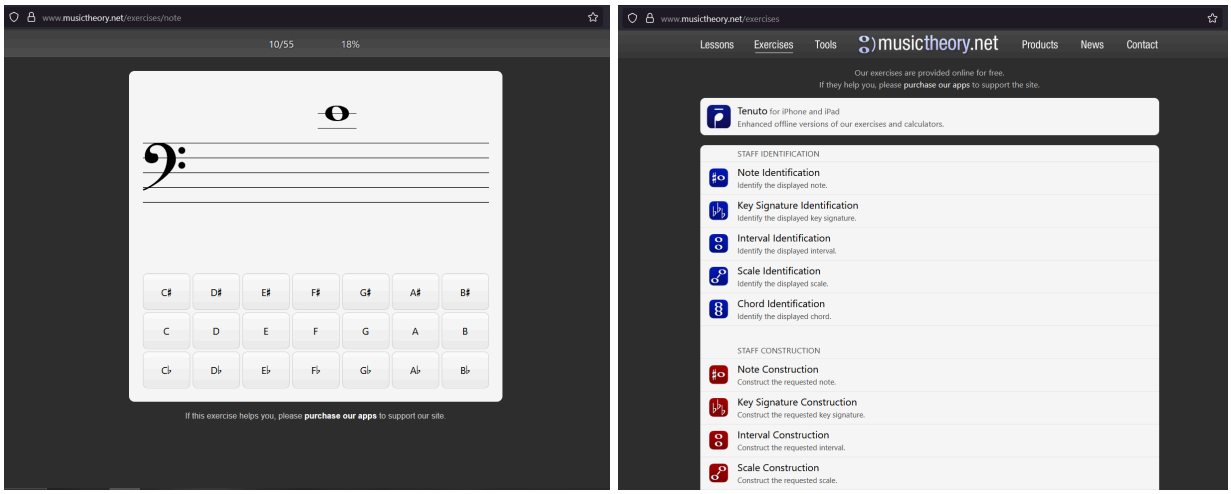

Figure 2: MusicTheory's User Interface, Web-Based

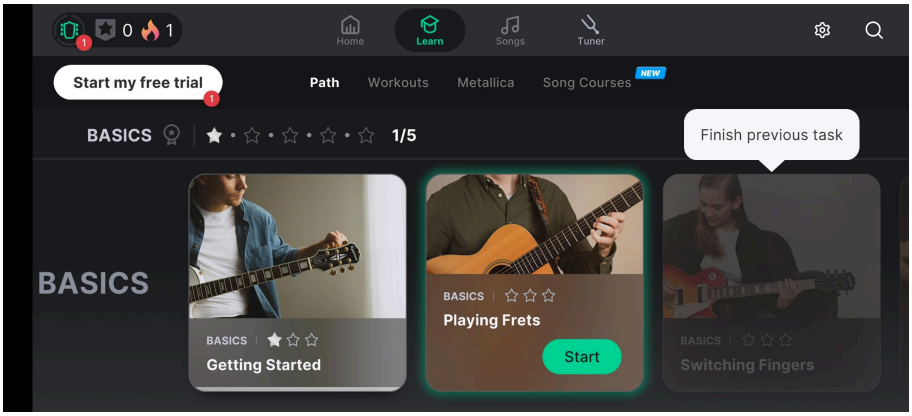

Figure 3: Yousician's User Interface. Strictly Horizontal Layout.

Before beginning development on the application, we did some preliminary research in order to get an idea of the general perception and success of AI use in education, as well as methods used in applying artificial intelligence in such environments. Ce Song explores the effect of using AI tutors and tools in music education, finding a positive correlation between "technology acceptance and reduced cognitive load, increased well-being and improved academic success." The research used AI tutors that followed the pace of the individual students and adapted to the most effective way to teach a specific student. This personalized tutor shows promise in improving the academic support that could help beginner / intermediate players get into learning how to play guitar. This supports our research and gives precedent for our AI powered application to be of benefit to our users (Ce).

Other articles focused on the actual methods used in integrating AI, such as Lauri Väkevä and Heidi Partti's, in which they demonstrate the different ways that Generative AI can be used to learn music theory. This is done by utilizing it for the creative aspects of music rather than the technical. The article also shares the positives, negatives, and uncertain future in music education in general. Overall, the paper analyzes Generative AI's usefulness and where the use of it can remove agency from the user (Lauri). Furthermore, Jacob Holster explains the different ways in which a Generative AI model such as ChatGPT could be used in music education. It focuses on the simple chatbot model to take into consideration the interests of the student and the goals of the teacher to create a personalized and effective lesson plan. It also provides some information on how the instructor would structure prompts to get the best response from the AI. This is what we would take inspiration from for our application (Jacob).

We also looked at some articles to see the current challenges of AI use in education as well. Chris Rodes focuses on the development of an AI guitar assistant to provide a good learning experience to those who may not have the financial resources to get an instrument or lessons, highlighting the divide in learning capabilities in guitar given an individual's economic situation.

While this is a positive to those without the funds for a real music teacher, it does raise questions on the occupation of being a music teacher. Additionally, the article mentioned “Humanizing” the AI’s integration to be a good public application and mimic that of a tutor or friend. Moreover, it explores the gamification of learning when using AI to incentivize students who use the AI to promote engagement (Chris). In the literature review article, Mycka presents the current research on music education and artificial intelligence and its effect on both students and educators. It also explores the methods used to create AI models and the issues with music generation due to the inherently creative nature that composition brings. It is mostly a compilation of the many aspects of music that have not been ironed out when it comes to AI integration. However, most of the issues with artificial intelligence in music stem from composition, which is not what this application will focus on (Mycka).

System Design

Our Application’s front end is developed with Next.js and React.js to accommodate client-side interactions and requests. For styling, we use Tailwind CSS and Shadcn UI for components across pages. On the backend, Strumii integrates Supabase, a cloud platform that prepares basic backend requirements for developers (like authentication, API calls). Our backend also uses PostgreSQL as its database management system and Prisma for our ORM (Object-Related Mapper) as a bridge between the backend server and database. We also use Docker to maintain the entire project and ensure compatibility across various systems that wish to use Strumii.

Given information from the user’s profile, we will use an AI model to generate a chord progression, varying from different genres, lengths, and difficulty levels. We are using Essentia JS’s pitch and chord detection to evaluate the accuracy of a user’s playing. Upon completion of a lesson, they will provide feedback to the model to generate an improved lesson.

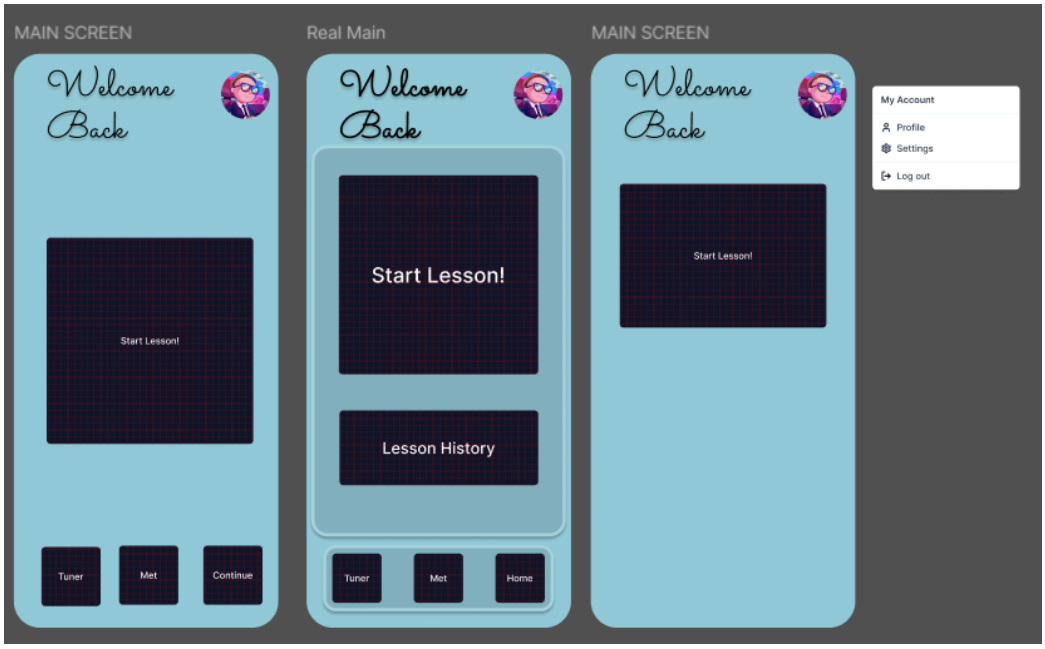

Figure 4: Initial Mobile Mockups

After logging into Strumii, the user would be prompted to either start a lesson, view their lesson history, or access other features like the tuner and metronome. While we were not able to complete these features given the time constraints, the general layout remained the same.

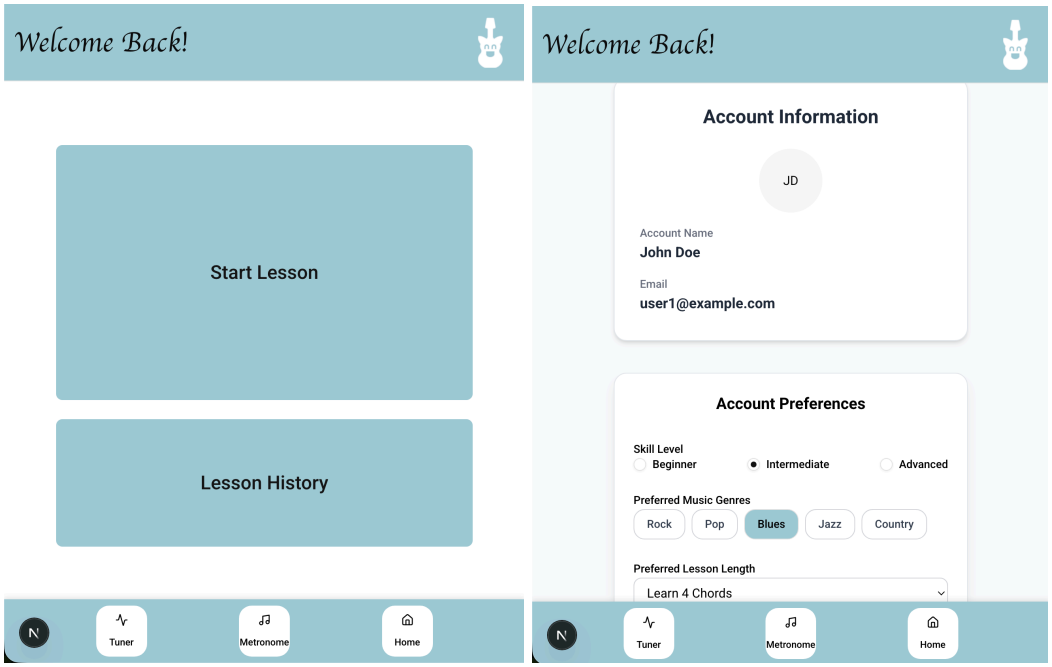

Figure 5: Final Mobile Mockups

The most noticeable changes are the color scheme and account information view, where users are able to set account preferences to match their skill level, preferred music genres, and the length of their lesson.

Figure 6: Initial lesson overview and feedback mockups

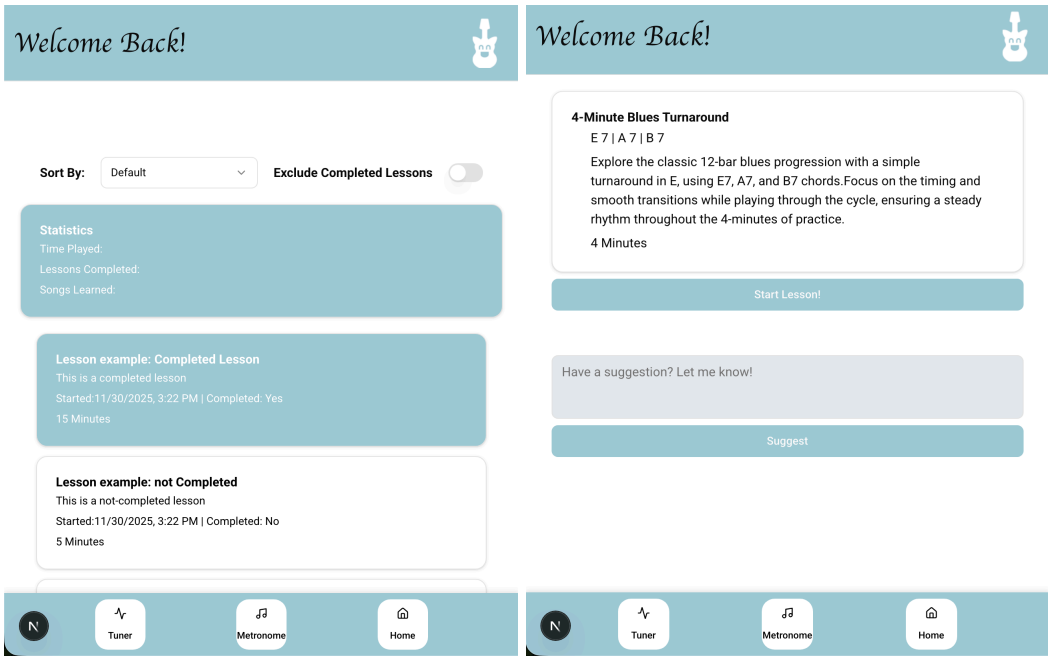

Figure 7: Final lesson overview page

Our lesson history page allows users to see the title of the lesson, the chord progression generated, and a short description of the lesson and expected duration. There are placeholders for statistics about the lessons, but they were not completed during the duration of the study. We also did not get to complete the user feedback page in Figure 6, but the concept was to provide feedback on the lesson, for instance, whether the chord progression did not match the preferred length or skill level, and have the application generate an improved lesson.

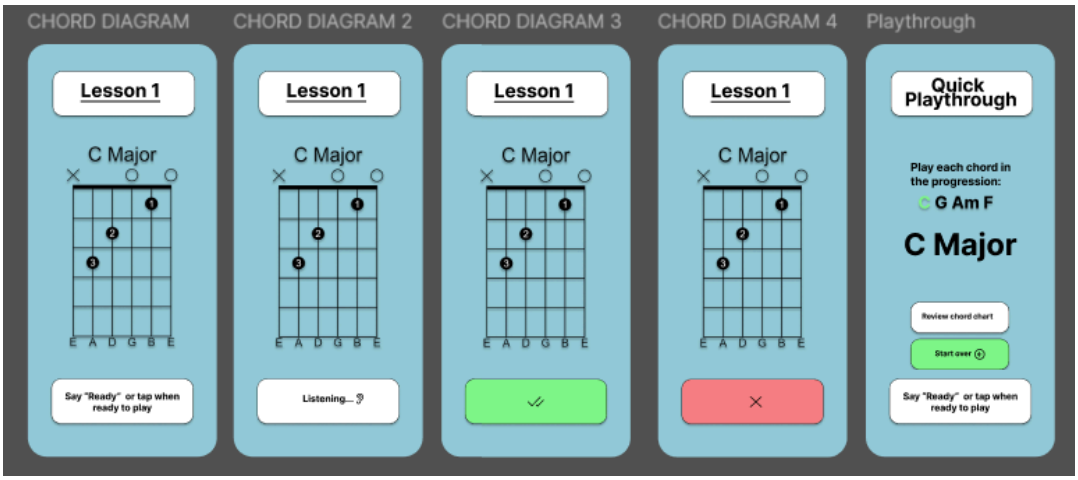

Figure 8: Initial lesson progression mockup

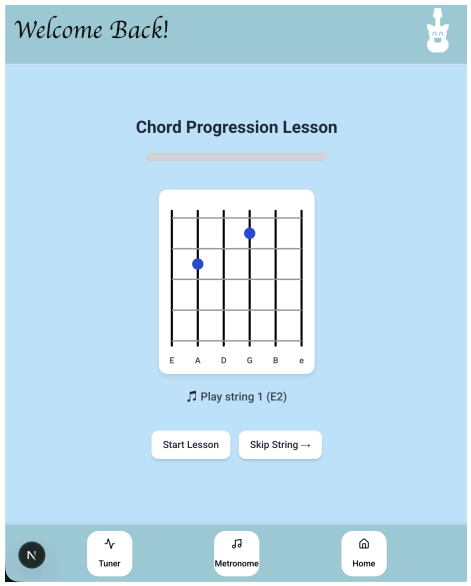

Figure 9: Final lesson progression page

The chord progression lesson page remained similar to the mockups, with the main difference being that the final product did not integrate the speech-to-text functionality. The user would play a string, then the pitch detector would recognize if that string is close to the correct note's

frequency. Finally, the detector will listen to a complete strum of all the notes and decide if they match the notes to move on to the next chord. For testing purposes, we added a skip string button to simulate this process.

Experiment Design

We want each participant in our study to evaluate Strumii's effectiveness through completing a pre-survey to collect demographic information through Google Forms and to assess their skill level and motivation. Then, a participant will complete a sample lesson guided by the development team in person. Upon completion of the lessons, they share any feedback or improvements to the app, as well as any feelings regarding the use of AI-generated chord progression lessons.

Figure 10: Experiment flow

The survey questions covered basic demographics like age, gender, and college status, as well as guitar-related experience, including how long participants have played guitar, how often they practice, and their comfort with fundamental skills like open chords, chord transitions, simple melodies, and music theory concepts. Participants were also asked about the specific skills they want to improve and the styles of music they are most interested in learning. Finally, the last few survey questions were focused on AI usage, like how often participants use AI tools, their trust in these tools, and whether or not they already use them for academic purposes. Through the survey, we can understand the participants' needs both for music education and how to construct AI-driven learning.

Then, one participant will use the app in person with the guidance of the development team to give their feedback. Based on the feedback, we will focus on improving the flow and usability of the lessons. They suggested that clicking through each step was "tedious and takes the focus off actually learning the chords," so we will keep this in mind. However, there were also qualities that the user said they enjoyed, like "being able to regenerate the lesson and easily update their preferences."

Results

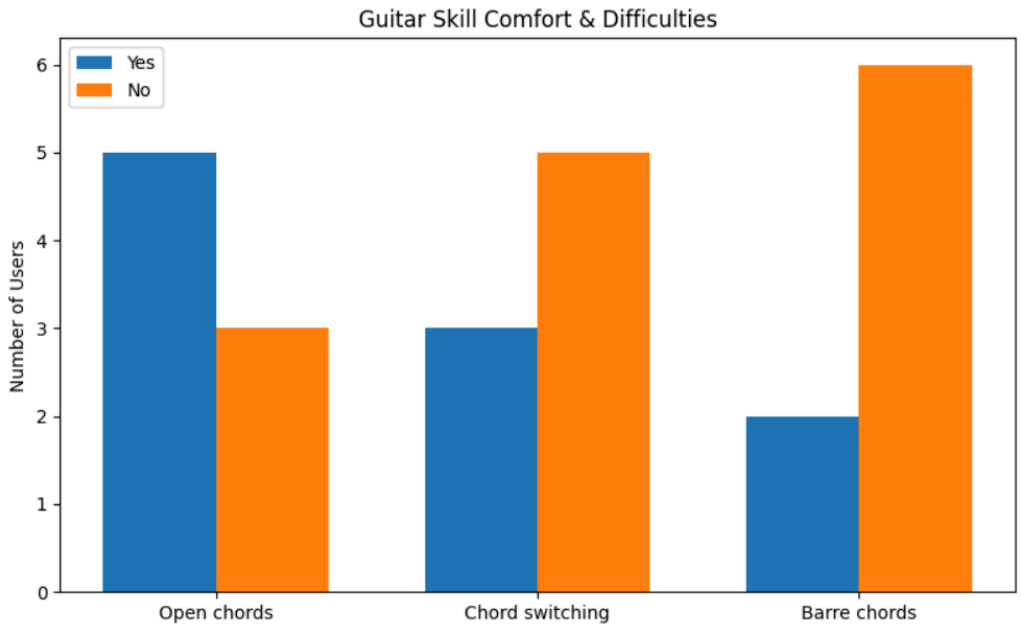

Figure 11: Guitar skill & difficulties survey result

Figure 11 summarizes the participants' comfort with core guitar skills. While the majority of users reported being able to play basic open chords, even fewer were comfortable with chord transitions and barre chords. This suggests that there is a gay between foundational skills and practical execution. With Strumii, we want to close this gap as users begin to practice switching between chords. While barre chords are also a point of difficulty among our users, we believe it may be naive to include them in our lessons, so we decided to focus more on transitions between open chords.

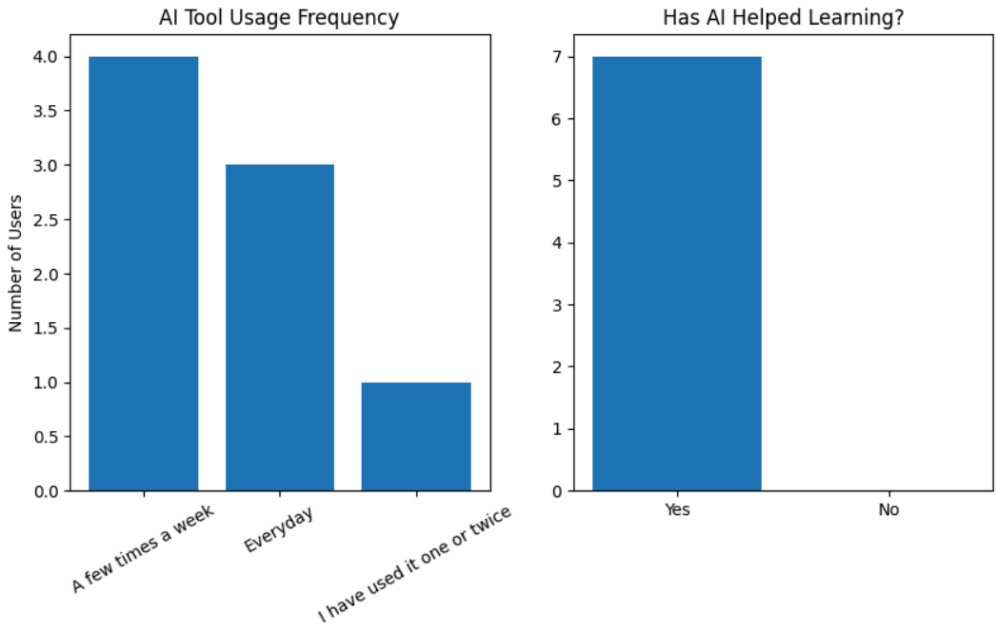

Figure 12: AI usage survey result

Figure 12 presents participants' engagement with AI tools and their perceived impact on learning. Most respondents reported using AI tools nearly every day or a few times a week, indicating high familiarity with AI in general. Additionally, nearly all the participants indicated that AI has helped them understand concepts they once found difficult. These results indicate that users are both comfortable with AI tools and are open to using them as a resource to learn, which is the motivation for Strumii.

Discussion

There is a gap in the literature regarding the use of artificial intelligence in musical instrument learning due to the lack of research on their intersection. Even though existing approaches focus on static lessons, participants could benefit from using their familiarity with AI to learn through personalized lessons as if simulated by a real instructor. We believe that this can serve as motivation for continuing music education, since users are able to tailor their learning to their own preferences. Strumii partially addresses this gap through its regenerative lessons and dynamic user preferences.

Due to time constraints, we were unable to do official participant testing and feedback for more users. Ideally, we would have been able to collect more data about how the participants engage with the app. These would be details such as the frequency of starting a lesson through a log-in timestamp, completion of a lesson, and the duration of time spent learning on the application. Moreover, it would be helpful to devise a way to translate the lesson history data into a format that the LLM would be able to understand and incorporate into lesson generation. Then, we would be able to further improve the adaptability of the lessons, given that the model can

understand and learn from the user's previous lessons. It would also be helpful to observe the users engaging with the app to gauge their enthusiasm and interaction with the AI elements of the app.

Conclusion

Our study aimed to observe if using artificial intelligence to generate personalized lessons would improve their understanding of the fundamentals of guitar. This was accomplished by collecting user preferences and incorporating them into the prompting to create lessons that the user would be more interested in. Previously, there have been similar music apps that incorporate AI into their apps. They each have their own drawbacks, some from usability issues, time constraints, and price.

Although this research study was limited in its data collection, there are still improvements to be learned from it. The data that we managed to record mainly came from our initial survey of participants about their previous experiences with music learning and artificial intelligence. Our post-experiment data resulted from a participant interview after using the most developed version of Strumii. Using both sources of user information, we learned that AI has the potential to support users in achieving their learning goals by providing personalized lessons and guided practice. AI-driven applications like Strumii can meaningfully assist students in overcoming the constraints of in-person lessons.

With this information and participant feedback, we can continue to improve the user experience in order to further our understanding of how students are interacting with artificial intelligence in a learning capacity. In the future, we hope to further determine the effectiveness of artificial intelligence's ability to create personalized learning.

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